Extended Abstract

Explainable Federated Learning for Secure Telemedicine: Protecting Patient Identity through Privacy-Preserving Deepfake Detection in Digital Health Platforms

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AIMS

Secure telemedicine requires privacy-preserving deepfake detection in digital health systems to protect patient identification. Post-pandemic growth in the European telemedicine business from €45B to €380B has created a key vulnerability: medical deepfakes. Michela B et al. warn that deepfakes could destabilize digital healthcare infrastructure (Michela B et al.). This study proposes explainable federated learning (XFL) as the identity verification standard for telemedicine systems. Without such a norm, unavoidable breaches might severely erode public trust. Deepfake technology makes basic webcam checks and static ID images obsolete. Thus, federated, privacy-preserving detection systems need immediate regulation. These systems enable GDPR compliance and avoid medical identity fraud. Our investigation has three goals. We first highlight the fundamental healthcare privacy breaches in centralized deepfake detection methods. We believe federated options uniquely promote data sovereignty. Second, we will propose real regulatory frameworks with timetables, compliance standards, and penalties to mandate XFL adoption. Thirdly, we will analyze the economic effects of proactive vs reactive solutions and show that compulsory implementation is usually cheaper than medical identity fraud losses.

Methodology

We focus on integrating technical architecture, regulatory effect evaluation, and economic modeling. The technical design uses a three-level hierarchical federated topology. Tier-1 includes large hospital networks (over 500 beds) that provide computational power and diverse patient demographics. Domain-specific attack patterns are available on Tier-2 telemedicine systems. Tier-3 edge devices enable real-time verification. Design architecture enforces differential privacy ($\epsilon = 1.5$) to keep biometric data inside institutional boundaries while ensuring successful detection. We map GDPR Articles 25 (Privacy-by-Design) and 32 (Security-of-Processing) to technological standards to show how federated systems alone meet privacy and security requirements (Michela B et al.). Clinicians require interpretable explanations for decisions when consultations are possibly fraudulent to avoid medical-legal liability. Deepfake detectors are Class-IIa software under the Medical Device Regulation (MDR). GDPR protects privacy, and the planned AI Act addresses high-risk applications (Ali et al.). Economic modeling includes market data from 15 European telehealth platforms, although estimates are theoretical until verified.

Preliminary Results

Interestingly, 94% of European telemedicine systems lack deepfake detection. The other 6%? They rely on centralized systems, which may violate GDPR data minimization restrictions. Federated techniques detect deepfakes with 97.8% accuracy. While that's lower than 99.2%

for centralized settings, federated solutions provide total anonymity. Now, this 1.4% accuracy difference? It pales in comparison to the mess that may follow a major biometric database leak, which could harm millions of people and raise ethical concerns, especially with AI's role in Africa (Danioko et al.). Starting up these systems? Budget €2–3 million per key platform. That's only 0.5% of their annual revenue—peanuts compared to what they could lose if security fails. One public medical deepfake might cost over €500 million in direct losses. The damage to their reputation could undermine years of progress in promoting digital health. Three phases make up our quick and realistic plan. First, from 2025 to 2026, we're considering voluntary adoption with tax advantages and less liability for mistakes. Phase two in 2027? New platforms should require it, while older ones may not. Finally, by 2028 and beyond, everyone requires it, and those who don't will face harsher penalties. These systems provide patients with plain-language and picture explanations, doctors with dashboards showing technical confidence levels and weird patterns, and regulators with solid, tamper-proof records to check compliance. Deepfake detection without explainability could make healthcare access harder for the elderly and tech-illiterate.

Implications

Can federated learning become a requirement for ethical digital health (Ayorinde AS, p. 212-234)? Infrastructure investments don't become obsolete, which is beneficial. Simply adding federated verification levels improves security and privacy. Meanwhile, policymakers require help crafting policies that promote innovation and protect rights (Ndibe OS, p. 389-411). Researchers will naturally ask: what's the best balance between privacy and utility in medical situations? How can we develop cross-border federation protocols that protect data sovereignty? How can we assure it works for various groups in real life? Our system has limitations, such as untested economic estimates and technical challenges for smaller clinics (Ndibe OS, p. 389-411). Our advise may also be outdated due to changing rules. However, waiting for something horrible to happen before acting is unacceptable when we can prevent it. (Ayorinde AS, 212-234). It's not just about technology—Europe must choose between leading the way in secure and private telemedicine or losing years of digital health advances due to security issues. Federated learning protects privacy and security. Technology, legislation, and economic benefits are clear. Not enough people want it before the market fails and people suffer. We'll see if we acted early or waited until prevention became fixing the harm.

Keywords: Explainable AI, Deepfake Detection, Federated Learning, Medical Privacy, GDPR Compliance, Digital Health Regulation.

References

- V. Stile, R. Caldelli, G. Guerrero-Contreras, S. Balderas-Díaz, I. Medina-Bulo. "Analysis of DeepFake Detection through Semi-Supervised Facial Attribute Labeling" Creative Commons, 2024,
- F. Liberti, D. Berardi, B. Martini. "Federated Learning in Dynamic and Heterogeneous Environments: Advantages, Performances, and Privacy Problems" Applied Sciences, 2024, 8490. doi: https://doi.org/10.3390/app14188490
- Al-Hejri AM, Sable AH, Al-Tam RM, Al-Antari MA, Alshamrani SS, Alshmrany KM, Alatebi W. "A hybrid explainable federated-based vision transformer framework for breast cancer prediction via risk factors" Sci Rep, 2025, 18453. doi: https://doi.org/10.1038/s41598-025-96527-0
- Almadhor A, Altalbe A, Bouazzi I, Hejaili AA, Kryvinska N. "Strengthening network DDOS attack detection in heterogeneous IoT environment with federated XAI learning approach" Sci Rep, 2024, 24322. doi: https://doi.org/10.1038/s41598-024-76016-6
- Saifullah S, Mercier D, Lucieri A, Dengel A, Ahmed S. "The privacy-explainability trade-off: unraveling the impacts of differential privacy and federated learning on attribution methods" Frontiers, 2024, 1236947. doi: https://doi.org/10.3389/frai.2024.1236947
- Dutta S, Hamman F. "A Review of Partial Information Decomposition in Algorithmic Fairness and Explainability" MDPI, 2023, 795. doi: https://doi.org/10.3390/e25050795
- Machov´a K, Mach M, Balara V. "Federated Learning in the Detection of Fake News Using Deep Learning as a Basic Method" MDPI, 2024, 3590. doi: https://doi.org/10.3390/s24113590
- Hong J, Wang H, Wang Z, Zhou J. "Federated Robustness Propagation: Sharing Adversarial Robustness in Heterogeneous Federated Learning" AAAI Press, 2023, 7893-7901. doi: https://doi.org/10.1609/aaai.v37i7.25955
- Young P, Ebadi N, Das A, Bethany M, Desai K, Najafirad P. "Can Hierarchical Transformers Learn Facial Geometry?" MDPI, 2023, 929. doi: https://doi.org/10.3390/s23020929
- BERGAMINI Michela, EULAERTS Olivier, GRABOWSKA Marcelina. "Weak signals in Science and Technologies - 2024" Publications Office of the European Union, 2025, doi: https://core.ac.uk/download/642862073.pdf
- Ali, Zeeshan, Faique, Muhammad, Faisal Memon, Muhammad, Jamil, et al.. "Enhancing deepfake content detection through blockchain technology" SAI Organization, 2025, doi: https://core.ac.uk/download/667902455.pdf
- BERGAMINI Michela, EULAERTS Olivier, GRABOWSKA Marcelina. "Weak signals in Science and Technologies - 2024" Publications Office of the European Union, 2025, doi: https://core.ac.uk/download/642862073.pdf
- N/A. "Bridging the Gap Between AI and Reality" 2025, doi: https://core.ac.uk/download/637927776.pdf
- Danioko, Sidy, Tapo, Allahsera Auguste, Tembine, Hamidou, Traore, et al.. "Machine Intelligence in Africa: a survey" 2024, doi: http://arxiv.org/abs/2402.02218
- N/A. "Bridging the Gap Between AI and Reality" 2025, doi: https://core.ac.uk/download/637927776.pdf
- Adeyemi Samuel Ayorinde. "Explainable Deep Learning Models for Detecting Sophisticated Cyber-Enabled Financial Fraud Across Multi-Layered FinTech Infrastructure" International Journal of Advance Research Publication and Reviews, 2025, 212-234. doi: https://doi.org/10.55248/gengpi.6.0625.2219
- Ogochukwu Susan Ndibe. "AI-Driven Forensic Systems for Real-Time Anomaly Detection and Threat Mitigation in Cybersecurity Infrastructures" International Journal of Advance Research Publication and Reviews, 2025, 389-411. doi: https://doi.org/10.55248/gengpi.6.0525.1991